

Diffusivity of ethylammonium nitrate protic ionic liquid confined in porous glasses

Andrei Filippov^{a,*}, Oleg N. Antzutkin^a, Victor P. Arkhipov^b, Oleg I. Gnezdilov^c

^a Chemistry of Interfaces, Luleå University of Technology, SE-971 87 Luleå, Sweden

^b Department of Physics, Kazan National Research Technological University, 420015 Kazan, Russia

^c Institute of Physics, Kazan Federal University, 420008 Kazan, Russia

ARTICLE INFO

Article history:

Received 11 January 2022

Revised 19 February 2022

Accepted 22 March 2022

Available online 29 March 2022

Keywords:

Ionic liquid

Self-diffusion

Tortuosity

Ion dynamics

Interaction with pores

ABSTRACT

We studied the diffusion of ionic liquid ethylammonium nitrate confined within pores of two types of porous glass, Vycor and Varapor, with average pore sizes of 4 nm and 9.8 nm, respectively, by ¹H NMR in the temperature range of 295–325 K. The diffusional behavior of the ionic liquid corresponds to long-term diffusion in a system of interconnected pores. It was shown that the diffusivity of EAN confined in Varapor is controlled by the porous system's tortuosity and does not need to take into consideration the interaction with the surface of the discrete pore walls. In the case of Vycor, the long-term diffusivity is a factor 1.5 lower than that expected in the absence of interaction with the pore walls. Two possible mechanisms that may explain this discrepancy are the EAN-surface interaction and retardation of EAN diffusion compared to *n*-decane in smaller pores present in Vycor porous glass due to pore size distribution. Confinement of EAN in nanoporous glass does not lead to the EAN phase transformation observed earlier for alkylammonium nitrates enclosed in a micrometer-sized layer. Prolonged exposure of EAN to a strong static magnetic field does not lead to changes in the NMR and diffusivity of EAN over time.

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1. Introduction

Ionic liquids (ILs) have attracted much interest as solvents for chemical synthesis [1], components of lubricants [2] and electrolytes [3]. For many of its practical applications such as in mechanical devices, supercapacitors, batteries, and catalytic reactors, etc. [1–7], understanding the properties of ILs close to a solid surface or in confinement has crucial importance. The properties of ILs at the liquid/surface interface and in confinement can be significantly different from those in bulk: they are affected by the size, shape and topology of geometrical restrictions. In particular, understanding the dynamic properties of ions is essential due to their relationship to chemical reaction rates, electrical conductivity and viscosity. The diffusivity of ionic liquids near interfaces and in confinement is an important topic of study [6–14].

Ethylammonium nitrate (EAN) is one of the most frequently studied ILs [15,16]. EAN has three readily exchangeable protons on the –NH₃ group of cations; therefore, it belongs to a class of so-called “protic” ionic liquids [15]. EAN is widely used as a reaction medium, a precipitating agent for protein separation [17], and an electrically conductive solvent and electrolyte for superca-

pacitors [18,19]. Therefore, the behavior of EAN near solid surfaces and in confinement has attracted special interest during the last few years [8,12–14,20,21]. The properties of EAN near surfaces and in confinement differ from those in bulk. Experimental AFM data have been reported using the approach of two hydrophobic mica surfaces separated by EAN, which revealed long-range (up to two nm) forces that appear to be electrostatic in origin [20]. A number of solvation layers of EAN was observed on highly-charged, smooth mica, while the lower surface charge of silica and increased roughness produces fewer layers [20]. NMR data showed that the hydrophobic surface of PETF induces regions of EAN in its vicinity, where diffusion of cations and anions is reduced compared to the bulk [14]. For confined EAN, a molecular dynamics simulation [12] showed that the diffusivity of ions between two walls, at distances from one nm to eight nm, decreases for very low slit sizes, recovering the bulk behavior as the distance between walls increases. The effect of the interface in prolonged ion dynamics only extends to a quite small region (ca. 1.35 nm) next to the walls. Overall behavior of the nanoconfined liquid in the slit pore can be described by the superposition of a bulk and a confined simulation with a fitted wall separation. The diffusivity of ethylammonium cation was studied for EAN layers enclosed between micrometer-spaced borosilicate and quartz glass plates [8,13,22]. It has been shown that the diffusion coefficients of cation is higher

* Corresponding author.

E-mail address: andrei.filippov@ltu.se (A. Filippov).

by a factor of ~ 2 – 3 compared with that in bulk [8]. It has also been found that both self-diffusion and NMR relaxation of the confined EAN reversibly alter after the sample is placed in a strong, static magnetic field [13,22].

The dynamics of protic ILs in nanopores has been studied by physicochemical methods. Nanoporous silica is the preferred confining matrix because of its easy synthesis [7]. Particularly, Vycor, a type of porous glass, is a good choice for this purpose. This material was extensively studied as a model system for the investigation of sorption and transport processes in porous solids and has various applications in chemical separation and catalysis [23]. The porous glass is manufactured in three stages [24]. First, a solution consisting of 75% SiO₂, 20% B₂O₃ and 5% Na₂O is quenched. During heat treatment, the substance separates into a phase rich in SiO₂ and a phase saturated with alkali metal oxide B₂O₃. The boron-rich phase is then removed by acid leaching, leaving an almost pure SiO₂ skeleton, with 96% silica and 4% boron trioxide. This leads to the formation of a random distribution of pores in space. The average pore diameter of Vycor is 4 nm, and the porosity of glass is about 30% [25]. In addition to the pore diameter, an important characteristic of glass is the chord length. The average length of the Vycor pore chord is ~ 7 – 9 nm [24,25], which is why the pores of this glass are believed to be predominantly cylindrical in shape. Vycor has a highly disordered structure with a network of pores containing areas of alternating expansion and contraction.

Nuclear magnetic resonance (NMR) is a unique, noninvasive method to study the diffusivity of molecules and different ions in opaque, porous media [10,26–28]. ¹H pulse-field-gradient (PFG) NMR covers a wide range of times ranging from milliseconds to seconds and corresponding lengths of molecular displacements in sub-micrometer and micrometer ranges. This technique provides an unprecedented opportunity to obtain detailed information about the relative permeability of porous media.

The purpose of this work was to study confined diffusion of ethylammonium nitrate in nanopores of Vycor and Varapor porous glass by ¹H NMR PFG to reveal basic features of the diffusion process.

2. Materials and methods

2.1. Materials

The structure of EAN is shown in Fig. 1. EAN consists of an ethylammonium (EA) cation and a nitrate anion. EAN was synthesized at the Chemistry of Interfaces, Luleå University of Technology. Synthesis of EAN follows the protocol of Evans *et al.* [29] and Poole *et al.* [30]. Before introducing the IL into pores, it was degassed under vacuum (pressure less than 10^{-3} mbar, temperature 313 K) for 60 h. The chemical composition of each sample (content of EAN and impurities) was analyzed by liquid ¹H, ¹³C and ¹⁵N

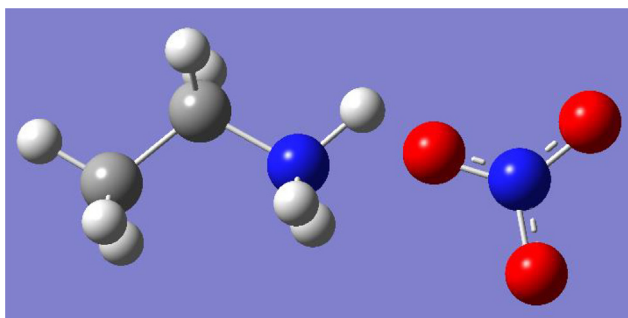


Fig. 1. Chemical structure of EAN: ethylammonium cation and nitrate anion. Nitrogen (blue), oxygen (red), carbon (dark gray) and hydrogen (light gray).

NMR. The ¹H and ¹³C NMR spectra demonstrate that the content of the ethylammonium protons and carbons were approximately 99%.

Vycor_7930 and Varapor_100 porous glass, both prepared by similar technology [24], were purchased from Advanced Glass & Ceramics (St. James, NC) [24,31]. Average pore diameters are 0.4 and 9.8 nm, respectively, and porosities are ~ 0.3 [24,31] and ~ 0.4 [32], respectively. More information about the porous glass can be found at <https://www.porousglass.com>. A sample of the porous glass has the form of a cylinder with a length of 15 mm and a diameter of 4 mm. Before use, each sample was cleaned from organic impurities by heating in a boiling 50% hydrogen peroxide aqueous solution at 393 K for 48 h. Afterwards, it was washed with distilled water and dried under vacuum. Activation of the glass was performed at 723 K for 2 h. It is known that Vycor type porous glasses contain about 96% of SiO₂ that resulting in their melting point close to that of pure quartz (1715 C). Therefore, the maximum operating temperature of Vycor is 1173 K (900 C) [33,34]. We used each sample of Vycor (and Varapor) several times in different experiments with cleaning and temperature activation afterwards. Porosity of the sample was checked each time after the activation procedure. Measurements showed that the treatment does not influence the porosity as well the diffusion data. The procedure for introducing EAN into nanopores is similar to that described earlier [7]. The glass cylinder was then cooled and placed under vacuum in a bath with EAN, then atmospheric pressure was applied for two days at 294 K. After that, the sample was removed from the bath, wiped with paper and placed in a glass sample tube. Just after preparation, the sample was heated at 330 K for 3 h to homogenize the distribution of the IL inside the pores. The sample was weighed and calculations taking into account the density of bulk EAN ($1.21 \text{ g}\cdot\text{cm}^{-3}$ at 298 K) showed that approximately all pores (29% and 40% of the sample volume for Vycor and Varapor, respectively) were filled. We have estimated a filling factor of the same porous glasses cylinders for water (H₂O) using bulk water density ($0.997 \text{ g}\cdot\text{cm}^{-3}$ at 293 K) and obtained the same filling percentages of the pores.

To compare the dynamics of ionic liquid EAN with the dynamics of a non-polar molecular liquid, a sample of Varapor saturated with *n*-decane (C₁₀H₂₂) was prepared in the same manner as for EAN.

2.2. NMR measurements

The ¹H NMR spectrum of confined EAN demonstrates three lines corresponding to NH₃ (6.65 ppm), CH₂ (2.32 ppm) and CH₃ (1.0 ppm) protons of the ethylammonium cation of EAN (Fig. S1 of the Electronic Supplementary Information, ESI). The spectrum did not change when the sample was kept in a magnetic field for up to several days. Pulsed gradient spin echo-nuclear magnetic resonance (PGSE-NMR) measurements were performed on a Bruker Avance III (Bruker BioSpin AG) NMR spectrometer. The measurements were performed on ¹H (400.21 MHz) with a PGSE-NMR probe Diff50 (Bruker). A sample was placed in a standard 5-mm glass sample tube. The tube was closed with a plastic stopper to avoid air contact. Prior to the measurements, the sample was equilibrated at a specific temperature for 15 min. The diffusional decays (DD) were recorded using the stimulated echo (StE) pulse train. For single-component diffusion, the form of the DD can be described as [28]:

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right) \exp(-\gamma^2 \delta^2 g^2 D t_d). \quad (1)$$

Here, A is the integral intensity of the NMR signal, τ is the time interval between first and second radiofrequency pulses, τ_1 is the time interval between second and third radiofrequency pulses. γ

is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta / 3)$ is the diffusion time; Δ is the time interval between two identical gradient pulses; and D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was $7 \mu\text{s}$, δ was in the range of $(0.5 \div 2) \text{ ms}$, τ was in the range of $(3 \div 5) \text{ ms}$, and g was varied from 0.06 up to the maximum of the gradient amplitude, $29.73 \text{ T}\cdot\text{m}^{-1}$. Diffusion time t_d was varied from 25 to 400 ms . The repetition time during accumulation of signal transients was 3.5 s .

There is a specificity in application of the NMR diffusion technique to study the diffusion of liquid confined in pores. It suggests that during the experimental diffusion time, the molecules were displaced by distances far exceeding the pore sizes. Thus, being orthogonal to the pore walls, the molecules undergo all possible orientations of the internal magnetic field gradients. Hence, in the monopolar version of the pulse sequence, Ste, the internal gradients contribute with some “average” effect. However, to demonstrate the absence of any effect of the internal field gradient directly, a stimulated echo sequence with bipolar gradient pulses [35] (SteBp) modified by including a longitudinal Eddy-current-delay [36] was also used. The latter pulse sequence removes cross-terms arising from the applied pulsed gradients, g and g_0 , which are significant even if g_0 values are small.

3. Results and discussion

^1H NMR diffusion decays in bulk EAN have a single-component form [8,14]. They represent diffusion of the ethylammonium (EA) cation, because the nitrate anion does not contain protons. The single-component form was also observed in EAN confined in pores. Examples of the DDs for EAN in pores of Vycor and Varapor are shown in Fig. 2.

Diffusion coefficients D s of (EA) were obtained from slopes of DDs. D s for confined EAN were not dependent on diffusion time in the studied range of 25 – 400 ms and were in the range of $5 \cdot 10^{-11}$ – $5 \cdot 10^{-10} \text{ m}^2/\text{s}$.

Temperature dependences of D for bulk and confined EAN are presented in Fig. 3 in Arrhenius plots at temperatures of 295 – 325 K . D increases with an increase of temperature for all three systems. For bulk EAN, our D is similar to that obtained by others [8,37,38]. It is seen that the dependences are quite linear in this temperature range. The slopes of the curves are close to each other and can be characterized by apparent energy of diffusion E_D according to the equation:

$$D = D_0 \cdot \exp(-E_D/RT). \quad (2)$$

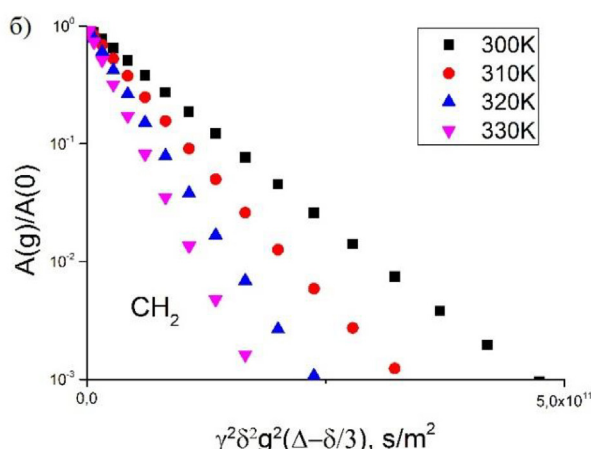
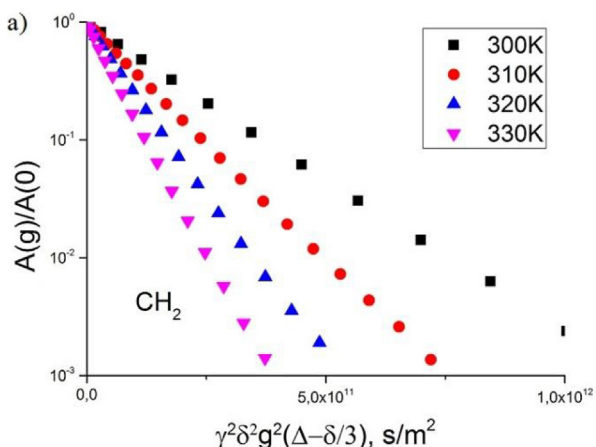


Fig. 2. ^1H diffusion decays of EAN confined in pores (pores: a) Vycor and b) Varapor.

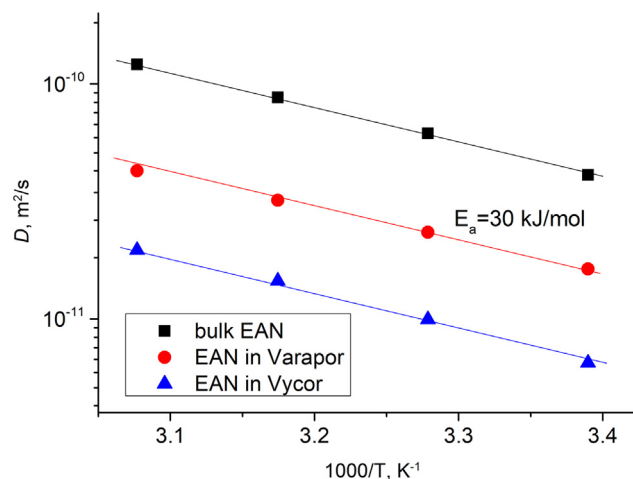


Fig. 3. Temperature dependences of diffusion coefficients for bulk EAN, and EAN confined in pores of Vycor and Varapor. Slopes of lines correspond to the apparent activation energy of diffusion 30 kJ/mol .

E_D for all temperature dependences are close to 30 kJ/mol . Values of D decrease as the pore size decreases by factors of ~ 0.34 in Varapor and ~ 0.13 in Vycor relative to the bulk value of D . The ratio of the diffusion coefficients in pores (~ 2.54) is inversely proportional to the ratio of the pore sizes (~ 2.45).

It is interesting to compare the observed effect of nanopores on the diffusion of the ionic liquid with that of a non-dissociated molecular liquid. We studied diffusion of n -decane confined in pores of Varapor at 100% filling. n -decane is a linear molecule with a diameter of 0.3 nm and length of 1.38 nm , which is not predisposed to an association. Diffusion decays were single-component in the three decimal order of the echo decay, similar to bulk and confined EAN (Fig. 2). The diffusion coefficient of the n -decane was not dependent on diffusion time in the studied range of 25 – 400 ms (Fig. S2). Values of diffusion coefficients for n -decane were lower by a factor of 0.34 than those in bulk at the same temperature. A decrease by the same factor for diffusivity in the pores was observed for EAN.

The diameter of the ion pair of EAN is around 0.58 nm [20], which is a factor of 7 and 18 smaller than pore diameters of Vycor and Varapor, respectively. The behavior of the average self-diffusion coefficient in a system of interconnected pores is determined by three factors [39], namely, (i) the self-diffusion coefficient of a pure liquid, (ii) the effect of interaction with the

surface of the pore walls, and (iii) the tortuosity of the pores [26]. The interaction of EAN with silica surfaces has been studied previously, both experimentally [20] and in simulation [12]. With both approaches, the formation of surface layers was found, ranging up to ~ 4 nm, with the gradient of diffusivity in the direction normal to the surface [12]. The diffusion coefficient is lower by a factor of five near the surface and gradually increases to that of the bulk value. Taking into consideration the sizes of the pores and the thickness of the layer (0.53 nm [20]), EAN could form up to seven surface layers in Vycor and up to eleven layers in Varapor. Therefore, the layers may occupy an essential fraction of the pore volume of Varapor and especially of Vycor. The contribution from the layers should provide a component with decreased diffusivity in diffusion decays. In fact, DDs are single-component (Fig. 2). This can be explained by several factors. Firstly, the roughness of the pore surface, as has been stated by Atkin and Warr [20], and, probably, the curvature of the surface, decreases the extension of layers. The roughness of the Vycor internal pore surface estimated from the transmission electron microscopy image [24] has an upper cutoff of less than 20 Å, which is similar to that of the flat surface used by Atkin and Warr [20], while the effect of the surface curvature has not yet been analyzed.

Secondly, there is an exchange of ions between the layers and the bulk liquid in the center of the pore, which leads to observation of an averaged diffusion coefficient during the diffusion time of the PFG NMR experiment. The value of diffusion coefficients inside discrete pores of Varapor and Vycor is unknown. There are also no experimental or literature data concerning the lifetime of ions in the EAN surface layers on silica. From our data, it is seen that the single diffusion coefficient is observed at diffusion time 25 ms. Therefore, the lifetime of ions in the surface layer is much less than this time. Diffusion times of our measurement correspond to the fast exchange regime when ions from layers and bulk change their locations many times demonstrating a single averaged diffusion coefficient.

There is also ion diffusion in the interconnected porous spaces of the porous glass types used here. Using Einstein's equation:

$$\langle r^2 \rangle^{0.5} \sim 2 \cdot D \cdot t_d, \quad (2)$$

values of measured diffusion coefficients and diffusion times, we can estimate a range of mean-squared displacement $\langle r^2 \rangle^{0.5}$ experienced by the EA cation in the pores. These values are in the range of ~ 1.6 – 20 μm , which much exceeds the pore sizes of Varapor and Vycor. Therefore, the averaging of diffusivities occurs inside discrete pores as well as in the scale of mean-squared displacement of EA during the diffusion time of the experiment.

To further understand the measured diffusivity of EAN in porous glass, we should discuss the general concept of fluid diffusion in a system of interconnected pores, which is usually very complicated. The behavior of the average self-diffusion coefficient of a fluid confined in an interconnected porous structure (a well-connected system), without specific interaction of molecules with the surface of the pore walls and phase transformations induced by the pores, indicates the existence of three diffusion time intervals [26,39]. In the (I) short-term range, when the root-mean-square displacements of liquid molecules are less than the sizes of linear pores, if there is no significant effect of the pore walls on the observed molecular diffusion coefficient, $D \sim D_0$. Otherwise, $D \sim D_{\text{pore}} < D_0$. In the time interval corresponding to (II) intermediate diffusion, the root-mean-square displacements of liquid molecules are comparable to the linear dimensions of the pores and the effective diffusion coefficient D^* decreases with increase of diffusion time in a manner determined by the pore system geometry. Finally, during (III) long-term diffusion, the root-mean-square displacements of molecules are larger than the correlation length of

the porous space ξ , and the motion of liquid molecules is averaged over the space of the system. For porous systems with a random structure, the diffusion decay is, as a rule, exponential and D^* is diffusion time independent, $D^* = D_\infty$. In a well-connected porous medium, D^* approaches at long times a finite value, reduced by a geometrical factor, known as the tortuosity α [26],

$$D^* D_0 / \alpha. \quad (3)$$

From the special scale of measurements, which much exceed the sizes of discrete pores as well as chord length, and also from the independence of diffusion coefficients on diffusion time, it is evident that our diffusivity measurements during long-term diffusion occur in an interconnected porous system [26,39]. This agrees with the results of ultra-small-angle neutron scattering measurements of dry and water-saturated Vycor [40], which showed that the correlation length of the porous structure of the porous glass is in the sub-micrometer range. In this case, a decrease in the measured averaged diffusion coefficients in pores of Varapor and Vycor relative to bulk EAN is due to both the interaction of EAN with the pore surface inside the discrete pore and by the tortuosity of the porous spaces. Strictly speaking, we cannot directly apply Eq.(3) to calculate tortuosity, because the D_{pore} is not equal to D_0 . However, we can take this ($D_{\text{pore}} = D_0$) as an approximation. Our calculation gives tortuosities of Varapor and Vycor as ~ 2.94 and ~ 7.69 , respectively. It is interesting to compare the effects of pores on the diffusivity of EAN with the effect offered by the same porous system on the diffusivity of non-charged molecules. We obtained diffusion data of the nonionic liquid *n*-decane confined in Varapor and obtained the value of tortuosity as ~ 2.94 , which is the same as that of confined EAN. Therefore, the interaction of EAN with pore surfaces does not contribute essentially to the long-term diffusivities of EAN in Varapor with relatively large pores. On the other hand, for *n*-decane confined in Vycor [39], $D^*/D_0 \sim 0.194$ that gives tortuosity for *n*-decane ~ 5.15 , which is less than that for EAN (~ 7.69). This apparent inconsistency can be explained by different rationale. Firstly, the increased tortuosity for the EAN-Vycor system in comparison with the tortuosity for *n*-decane can be due to the large $D_{\text{pore}} = D_0$ of EAN, which was taken for the estimation of tortuosity. 1) Indeed, if EAN interacts with pore surfaces and due to this D_{pore} is a factor of 1.5 less than D_0 , the tortuosity of Vycor for EAN is equal to that of decane. When confined in silica, the IL ions in contact with the wall interacts with Si-O and/or Si-OH groups [7]. 2) Another factor that may also increase the tortuosity of Vycor for EAN is the pore size distribution, which is always present in porous glass. Pore size distributions of Vycor have been analyzed previously by nitrogen adsorption, mercury intrusion [25], and by SAXS [41]. All studies demonstrated that an essential fraction of the pores have diameters less than one nm. An adsorption-desorption and Monte Carlo study [42] showed that Vycor has a highly disordered silica matrix with a network of pores containing alternating enlargements (voids) and constrictions ("necks") that determine the diffusion properties of Vycor. Additionally, diffusion particles are not only ions and ion pairs, but also large aggregates, which exist in bulk EAN [43] and may persist in the pore center. Such aggregates contain about 6–7 cations and anions, therefore scaling estimation gives an increase in diameter of diffusion particle by a factor of 1.91, up to 1.1 nm. Therefore, increases in apparent tortuosity of Vycor for EAN can be due to the decreased permeability of Vycor by a factor of 1.5 for the relatively large size of associated EAN in comparison with *n*-decane. The effect of pore size distribution is meaningful for smaller pores of Vycor, while does not play a serious role for larger pores of Varapor. The second suggestion is supported by the equality of activation energies of diffusion for bulk and confined EAN (Fig. 3), which means that the mechanism of diffusion is the same in bulk and in pores of Varapor and Vycor. Therefore, EAN in pores of the studied porous

glass may behave in the same manner as in bulk, in spite of its confinement. The absence of the effect of layering, which was expected from the AFM experiment [20] and computer simulation [12] for the interaction of EAN with a flat silica surface, may be a result of the roughness of the internal pore surfaces, their high curvature or any other factors that influence EAN layering. Both roughness and curvature could play a role to some extent in the long-term diffusivity of Vycor.

NMR has previously been applied to study dynamic properties of EAN in micrometer-sized layers enclosed between flat surfaces of glass and quartz [8,22]. It was shown that EAN placed in such confinement experienced a type of phase transition, while the newly formed phase is characterized by accelerated T_2 relaxation of $-NH_3$ protons and a factor of 2–3 enhanced diffusivity of EA cation. For EAN confined in porous glasses, the presence of nanoconfinement does not accelerate T_2 NMR relaxation of $-NH_3$ protons, as is seen from Fig. S1, where ratios of integral intensities of all spectrum lines are proportional to their content of protons. We cannot measure diffusivities inside discrete pores, but the similarity of long-term diffusion behavior of EAN and *n*-decane indirectly demonstrates that no acceleration of diffusion occurs inside discrete pores. Therefore, the new phase with accelerated ion dynamics in pores is not formed. A possible reason preventing the formation of the phase may be that the pore size is smaller than the critical nucleus of the new phase.

One more effect observed in micrometer-sized flat EAN layers is the reversible influence of a strong static magnetic field [13,22], which also leads to change T_2 NMR relaxation, EAN and Li^+ [44] diffusivity. For EAN confined in nano-porous glass, neither the 1H NMR spectra form nor the long-term diffusion coefficients change during the entire experimental time period, which was up to several days in certain cases. Therefore, the effect does not occur in EAN confined in nano-sized porous glass. This might be due to either the highly curved surfaces of discrete pores or the heterogeneous structure of the porous space. Magnetic field inhomogeneity due to a difference in susceptibilities of the solid support and the liquid may also play a role.

4. Conclusions

We studied the diffusion of ethylammonium cations of EAN in pores of porous glass, Vycor (4.0 nm diameter) and Varapor (9.8 nm diameter), by 1H NMR. The diffusional behavior of the ionic liquid corresponds to long-term diffusion in a system of interconnected pores. Comparison with the diffusion of *n*-decane in the same porous glasses showed that the diffusivity of EAN confined in Varapor is controlled by the porous system's tortuosity and does need to take into consideration interaction with the surface of the discrete pore walls. For EAN confined in Vycor, the long-term diffusivity is a factor of 1.5 lower than expected for an ion with no interaction with the pore walls. Two possible mechanisms that may explain this discrepancy are the IL-surface interaction and the retardation of EAN diffusion in comparison with *n*-decane in smaller pores present in Vycor porous glass due to proper pore-size distribution. Both these mechanisms may contribute to a certain extent to the observed decreased diffusivity of EAN confined in Vycor.

Confinement of EAN in nanoporous glasses does not lead to the EAN phase transformation previously observed for alkylammonium nitrates enclosed in a micrometer-sized layer, which was observed as an accelerated NMR relaxation of $-NH_3$ protons and enhanced diffusivity of the alkylammonium cation. Long-term exposure of EAN to a strong static magnetic field does not lead to changes in the NMR spectrum and diffusivity of EAN with time.

CRediT authorship contribution statement

Andrei Filippov: Conceptualization, Writing – original draft. **Oleg N. Antzutkin:** Validation, Writing – review & editing. **Victor P. Arkhipov:** Software, Data curation. **Oleg I. Gnezdilov:** Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

O.I.G. acknowledges the subsidy allocated to the Kazan Federal University (project number: 0671-2020-0051). "Scriptia Academic Editing" is acknowledged for English correction and proofreading of this manuscript.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2022.118998>.

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